

PCB123 For Easier And Faster PCB TURNOUT

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PCB123 is a software tool for printed circuit board (PCB) designing. Built with a robust feature set, with over 40-plus years of manufacturing, the tool helps you ease over the design process and get your PCB manufactured from their owners, Sunstone Circuits. Bundled with this month's DVD, we take a look at what this tool has to offer, in this article.

Increased rendering speed for a smoother user experience

With the latest release version 5.2.1, the developers have more effective graphics algorithms built-in, which makes the tool respond to the user's commands in real-time.

Doing away with the old method of re-calculating all parameters to reciprocate the user's action, these improved algorithms allow the tool to read polygons off the image file itself. Be it in moving the layout to a new location on the board, zooming in to see a particular part of the layout or scaling up, the tool re-populates almost instantaneously, reproducing the change(s) on the screen with very little delay.

A well-structured schematic editor

Every design is constructed on one or more schematic sheets, which can be laid out in a logical way and connected with ports. Alongside is the part library window, which contains libraries from which you can select components for your schematic design.

Selecting the right component is easy as hovering the mouse

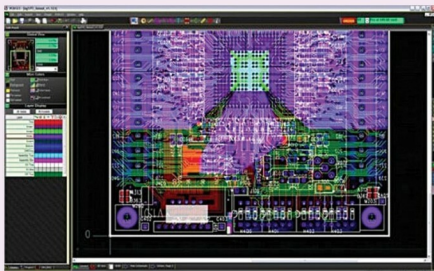


Fig. 1: Designing with PCB123 (Image courtesy: <http://xtronic.org>)

Kick-starting your design

A startup guides you with the initial configuration. The PCB layout you create in a program might turn out to be the base for many more designs you might do in the future, and so it might be wise to save the design template. The same can be loaded into a new design, on starting.

You can even choose from the already readily-provided templates or create a simple rectangle by specifying dimensions. The tool lets the user choose the number of layers on the board, solder mask and silkscreen options, PCB thickness and the weight of the copper deposits. The PCB can also be made lead-free if the user chooses to.

over each component displays a list of the properties defined for the selected component, including specified part number. There is also a symbol view that changes with every component selected—a very useful feature when part names are unknown.

A board layout that goes hand in hand

As the schematic is being created, the board stacks up on the footprints of the selected symbols. While these get stacked in a corner, the auto-place feature can be used to centre

these on the layout, resulting in what is called a rat's nest. This is basically a jumble of wires criss-crossing each other to their destination nodes, in keeping with the schematic. Routing between blocks can be done manually or with the help of the tool's auto-route feature.

Also, next to the layout are sections like different views, general tools and appearance, using which the user can work around the layout, concentrating on the areas that need detailed attention. From the manage parts tab, the user can add footprints to a newly-created multi-